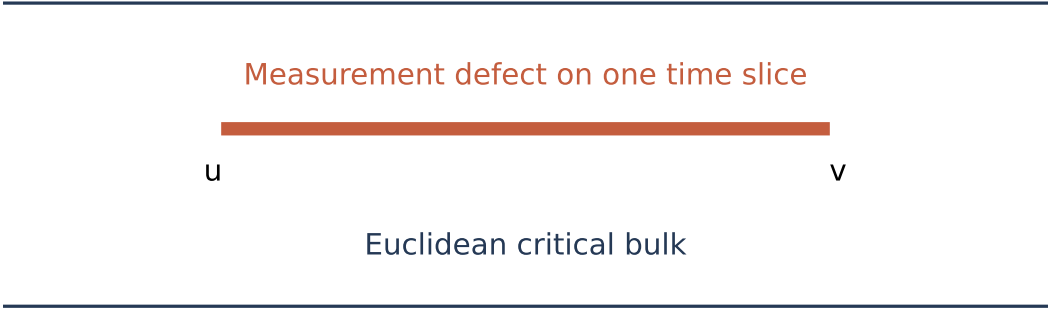A defect can change scaling without changing the bulk Hamiltonian.